10. Closest Pair of Points – Brute Force

AIM

To write a Python program to find the **closest pair of points** among a given set of points using the Brute Force algorithm by calculating the Euclidean distance between every pair of points.

PROCEDURE

1. Import the `math` module to calculate the Euclidean distance.
2. Define a function that accepts a list of points as input.
3. Initialize the minimum distance to infinity.
4. Use two nested `for` loops to compare every possible pair of points.
5. Calculate the Euclidean distance between each pair using the distance formula.
6. Compare the calculated distance with the minimum distance.
7. If a smaller distance is found, update the minimum distance and closest pair.
8. Repeat the process until all pairs of points are checked.
9. Display the closest pair of points and their minimum distance.

main.py | Visualize | Run ▶ | Output

```python
import math
def closestPair(points):
    min_distance = float('inf')
    closest_pair = None
    n = len(points)
    for i in range(n):
        for j in range(i + 1, n):
            x1, y1 = points[i]
            x2, y2 = points[j]
            distance = math.sqrt(
                (x2 - x1)**2 + (y2 - y1)**2
            )
            if distance < min_distance:
                min_distance = distance
                closest_pair = (points[i], points[j])
    return closest_pair, min_distance
points = [
    (2, 3),
    (12, 30),
    (40, 50),
    (5, 1),
    (12, 10)
```

```
Closest Pair: ((2, 3), (5, 1))
Minimum Distance: 3.61
```

RESULT

Thus, the Python program successfully identifies the **closest pair of points** and calculates the minimum Euclidean distance using the Brute Force algorithm.